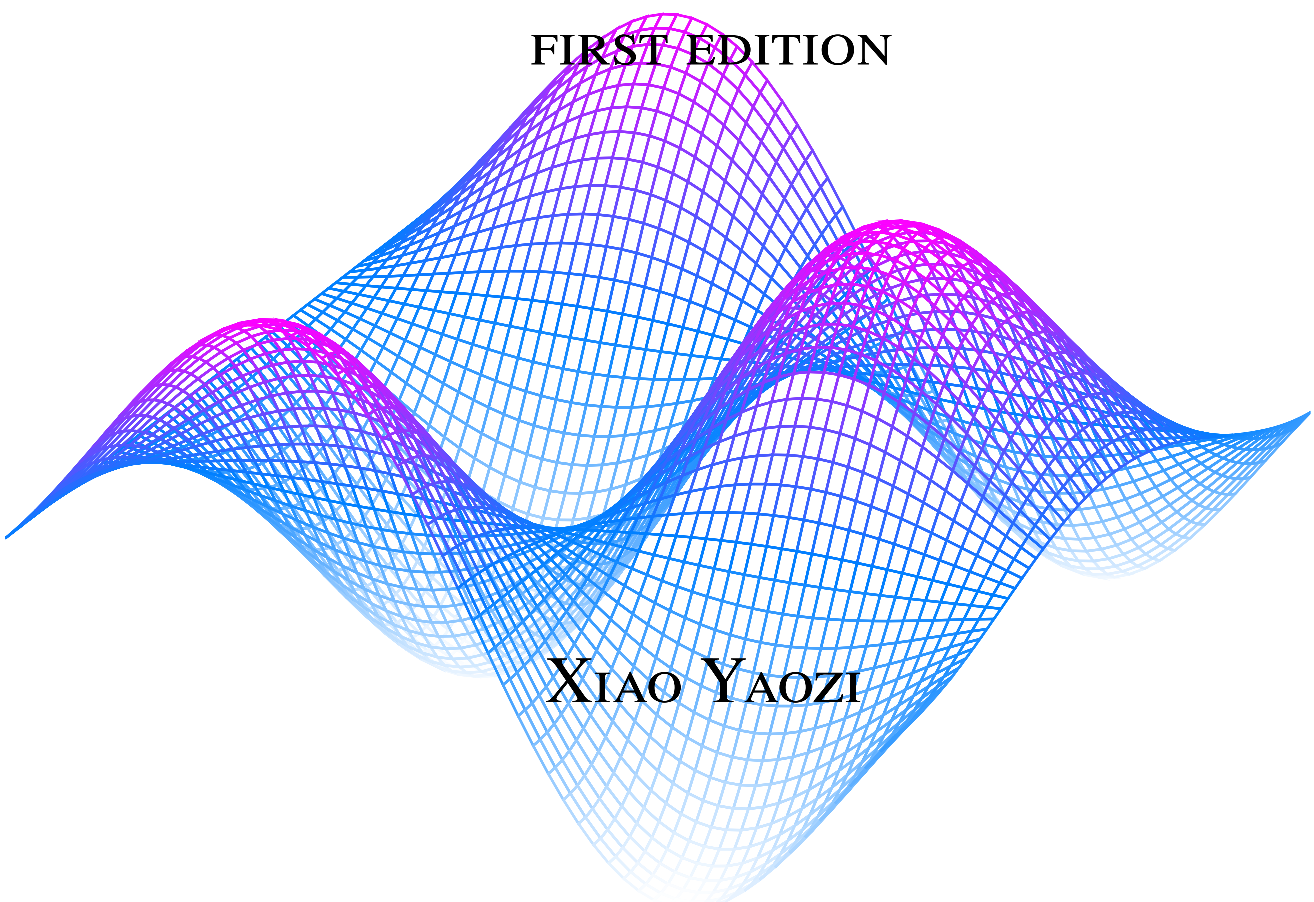
The Calculus Method

FIRST EDITION



Xiao Yaozi

Preface to the first edition

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Contents

PART I

Precalculus

CHAPTER 1

Functions

SECTION 1.1

Function basics

Before we start on the calculus topics, this part of the book will acquaint you with the mathematics you would need to understand in order to read this book. If you are familiar with the these topics, you may skip to the Part II on Calculus I. For the rest of you, the pace of this part will be very fast, it is recommended that you read up on the concepts you do not understand very well. We will start with the idea of a *function*. By now, I am sure that many of you may have seen equations like these:

$$x^2 - 5x + 6 = 0 \tag{1.1}$$

$$\sin(x) = 0.5, 0 \leq x \leq \frac{\pi}{2} \tag{1.2}$$

$$y = mx + c \tag{1.3}$$

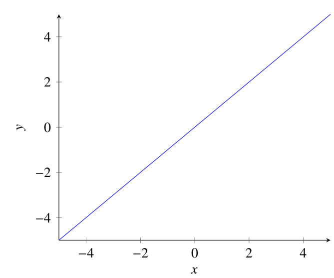


Figure 1.1. Graph of y=x.

Out of the three equations, notice that Equation (1.3) is special. There is not some fixed x values that you must plug in. Instead, the y value varies with the x value. Given a x value, the equation "produces" a y value. This brings us to the idea of a function. A function is simply something that takes in a value, and "produces" another value based on the input. You may have came across functions before, such as the trigonometric functions like sin and cos. The value of $\sin(x)$ varies with different values of x. Of course, it is time for me to give you the definition of a function.

Definition 1.1 A function f is a mathematical object that assigns its inputs, or *domain*, to its outputs, or *range*. The notation $f : A \to B$ means that the function has a domain A and range B. The notation $f(a)$ where $a \in A$ denotes the element of B to which a is assigned to.

Note that A and B are sets.

Example Here are some examples of functions:

$$f(x) = x + 1 \tag{1.4}$$

$$g(x) = \sin(x) \tag{1.5}$$

$$h(x) = x^2 - 5x + 6 \tag{1.6}$$

To evaluate the output of a function, you simply substitute the input to the expression that defines the function. Notice that for Equation (1.4), it is simply the line $y = x + 1$. Each value of $f(x)$ for some $x \in \mathbb{R}$ is just y. Take $x = 1$ for example, $f(1) = 1 + 1 = 2$ and for the line $y = x + 1$, notice the y value here is also 2. The function assigns each x value to $x + 1$. Hence, in this case, $f : \mathbb{R} \to \mathbb{R}$. For Equation (1.5) , the function takes any real x value and maps it to $\sin(x)$. As a result, its domain would be $\mathbb{R}$ and its range would be $[-1, 1]$. Again, it is just like the graph $y = \sin(x)$.

> Lastly, the quadratic in Equation (1.6) is another real function(i.e. it has a real domain and range). Think about what happens when $h(x) = 0$, what value of x must you plug in to make $h(x) = 0$. Well, isn't it just the roots of the quadratic. Hence, the roots of a function are just the values x such that $h(x) = 0$. In this case, it would be 2 and 3. Thus, $h(2) = h(3) = 0$.

Actually, functions can be more than the usual $f(x)$ = something. Let's say you want to have a function that outputs "1" if the input is rational and "0" if it is not, you can define it as such:

This is actually the famous Dirichlet function that shatters certain intuitions on functions and will be discussed in later chapters.

$$f(x) = \begin{cases} 1, & \text{if } x \in \mathbb{Q} \\ 0, & \text{if } x \notin \mathbb{Q} \end{cases} \tag{1.7}$$

Here, $f(\frac{1}{2})$ would be 1 but $f(\sqrt{2})$ would be 0. (You would probably get a migraine from graphing it.) This is an example of having conditionals in a function. Lastly, functions may take more than one input and produce more than one output, examples include:

$$f(x, y) = x^2 + y^2 \tag{1.8}$$

$$g(t) = (\cos(t), \sin(t)) \tag{1.9}$$

Equation (1.8) is an example of a *multivariate* function, in other words, it takes in multiple inputs. If $x = 2$ and $y = 2$, then $f(2, 2) = 8$. On the other hand, Equation (1.9) does not have a single output. Instead, it has a tuple as its output. Evaluating it at $t = \pi$, notice that the output is $(-1, 0)$. What is interesting is that if you trace out its outputs on a plane for all $t \in \mathbb{R}$, you will get:

As you will see in Part IV of the book, this is the parametric equation of a circle.

A tuple is basically a set where order matters. You can treat it as if it is a coordinate.

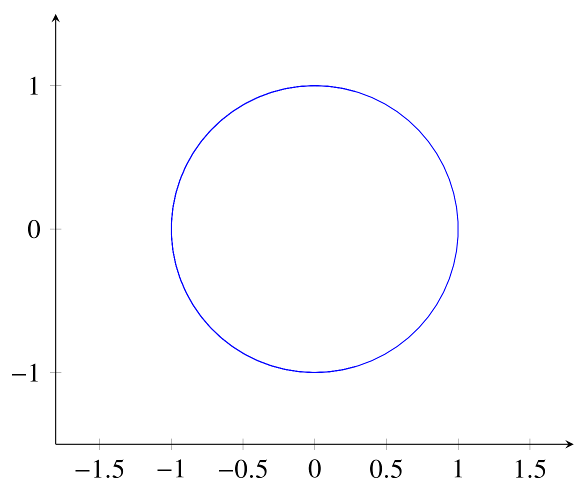


Figure 1.2. Plot of Equation (1.9) for all $t \in \mathbb{R}$

Functions are the building blocks of mathematics, and this book will deal extensively with it. Therefore, it is paramount that you get the basics right. Try the following exercise:

Exercise

1. Given that $x = 2$ and $y = \pi$, evaluate $f(x)$ for:
   (a) $f(x) = x^3 + 1$
   (b) $f(x) = 2^x$
   (c) $f(x) = \begin{cases} x^4 + 1 & \textit{if } x < 2 \\ 3^x & \textit{otherwise} \end{cases}$
   (d) $f(x, y) = \frac{\sin(y)}{x}$

2. Given the following functions, write down $f : A \to B$ where A is the domain and B is the range(do not consider complex inputs and outputs, we shall be ignorant to it for the time being to simplify things):
   (a) $f(x) = x$
   (b) $f(x) = \frac{1}{x}$
   (c) $f(x) = \sqrt{x}$

*This function has a special name, the **identity** function. Notice how its input and output are the same.*

3. Find the root(s) of the following functions:
   (a) $f(x) = x^2$
   (b) $f(x) = 2^x - 1$
   (c) $f(x) = \sin(x)$

Solution

1. (a) 9
   (b) 4
   (c) 9
   (d) 0
2. (a) $f : \mathbb{R} \to \mathbb{R}$
   (b) $f : \mathbb{R} \setminus \{0\} \to \mathbb{R} \setminus \{0\}$ (Note the function is not defined at $x = 0$)
   (c) $f : \mathbb{R}_0^+ \to \mathbb{R}_0^+$ (Square root not defined for negative numbers)
3. (a) 0
   (b) 1
   (c) $n\pi, \forall n \in \mathbb{Z}$

SECTION 1.2

Function composition and inverses

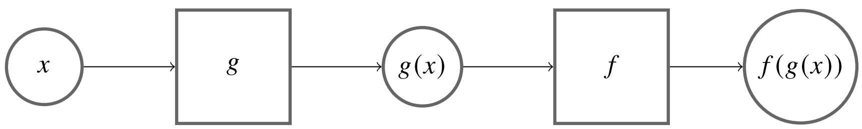
In this chapter, we will look at composition of functions and inverse functions. You can use the normal arithmetic operations on functions like how you would to numbers:

Example If $x = 2$ and $f(x) = x^2 + 4$:

$$f(x) + f(x) = 2^2 + 4 + 2^2 + 4 = 16$$

$$\frac{f(x)}{2} = \frac{2^2 + 4}{2} = 4$$

What is more interesting, perhaps, is taking the *output* of a function and use it as the *input* to another function. You may also think of it as "piping" one function into another. This is called *function composition*.



Note that the rectangles are functions and circles are values.

Figure 1.3. Visual representation of function composition.

Another notation for composing functions together(which we will not really use except for this chapter) is as such:

$$f(g(x)) = (f \circ g)(x)$$

When you compose the same function n-times, the notation for it is:

$$f^n(x) = \underbrace{f(f(\ldots f(x)))}_{n-times}$$

However, this does not apply to trigonometric functions. $\sin^2(x)$ *is* $\sin(x) \times \sin(x)$ *instead of* $\sin(\sin(x))$. *Annoying mathematical notation strikes again. What is even better, is that* $f^{-1}(x)$ *denotes the inverse of* f.

Example Here are some examples of composing functions together: If $x = 3$ and $f(x) = \pi x$, $g(x) = \sin(x) + 1$:

$$g(f(x)) = \sin(\pi \times 3) + 1 = 1 \tag{1.10}$$

$$f^3(x) = \pi^3 x \tag{1.11}$$

There is nothing much here, really, just evaluate the function from the innermost function to the outermost.

Theorem 1.1 Given functions $f : X \to Y$, $g : Y \to Z$ and $h : Z \to W$, prove that the composition of the three functions is *associative*, in other words:

$$h \circ (g \circ f(x)) = (h \circ g) \circ f(x)$$

PROOF Let $x \in X$, we have:

$$h \circ (g \circ f(x)) = h(g \circ f(x)) = h(g(f(x))) = h \circ g(f(x)) = (h \circ g)f(x)$$

This proves the associativity of function composition. QED

Remark This is just a fun little exercise for the reader and ultimately is not that important to the entire book, just a good identity to keep in the back of your mind.

Next, we will delve into inverse functions. Let us start with one I believe you may have seen before, look at the following figure:

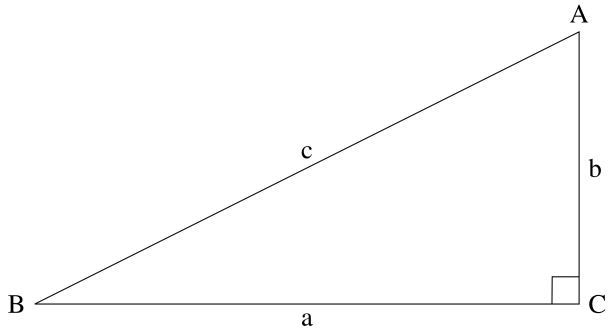


Figure 1.4. Figure of right triangle ABC.

If I ask you what is $\arcsin(B)$, you may say it is $\frac{b}{c}$. But, what is $\arcsin(\sin(B))$, well, that is just B, isn't it. Hence, the arcsin sort of "cancels" the sin. We call arcsin the *inverse function* or just inverse of sin. Generally, given a function $f : X \longrightarrow Y$, f^{-1} denotes the inverse function that satisfies:

We will use arcsin *as the inverse of* sin *instead of* $\sin^{-1}$ *as it is clearer.*

$$f \circ f^{-1}(y) = y, y \in Y \text{ and } f^{-1} \circ f(x) = x, x \in X \tag{1.12}$$

One common misconception is that $f(f^{-1}(x)) = f^{-1}(f(x))$. While this is true for many functions like $f(x) = x + 1$, it is *not* always true. For example, take the following function:

$$f(x) = \sin(x), f : \mathbb{R} \longrightarrow \{y : -1 \leq y \leq 1\}$$

This looks deceptively simple, when asked to evaluate $f^{-1}(f(\pi))$, it is π (You may want to verify it yourself). But what is $f(f^{-1}(\pi))$? Well, according to our definition of f, $f^{-1}(\pi)$ is simply not defined. In actuality, to find $f^{-1}(\pi)$, we need complex numbers, which is not allowed in the definition. Hence, you must be extra careful when you are dealing with function inverses and pay attention to the domain and range.

To find the inverse function of a function f, we have to find the function that maps the outputs of the f back to its inputs. To solve algebraically, we set $f(x)$ for some arbitrary x to y, and make x the subject of the equation.

Example

Given that $f(x) = \frac{x+7}{3}$:

$$\begin{aligned} f(x) &= y \\ \frac{x+7}{3} &= y \\ x + 7 &= 3y \\ x &= 3y - 7 \end{aligned}$$

Hence, the inverse function is $f^{-1}(y) = 3y - 7$.

Example

Some common examples of inverse functions:

1. Trigonometry:

   arcsin   arccos   arctan   ...

2. Logarithms (Inverses of exponential functions):

   ln $\log_b$

Exercise

1. Given $f(x) = \frac{x^2+3}{2}$ and $g(x) = \tan(x) + 1$, evaluate:
   (a) $f(g(\frac{\pi}{4}))$
   (b) $f^3(1)$
   (c) $f(g(\arcsin(0)))$
2. Given that $h(x) = (x-7)^2$, $x \geq 7$, find the inverse of h.
3. Hence, evaluate:
   (a) $h^{-1}(1)$
   (b) $h(h^{-1}(69))$ *As Peter Lorre would have said it, "Do it ze kveek vay, Johnny".*

Solution

1. (a) $\frac{7}{2}$
   (b) $\frac{61}{8}$
   (c) 2
2. $h^{-1}(y) = \sqrt{y} + 7$
3. (a) 8
   (b) 69 (Inverse function and the original function "cancels".)

SECTION 1.3

Injection, surjection and bijection

You may have noticed in the previous exercise that question 2 is particularly interesting. There is a restriction on the domain of $h(x)$. Why is that? This is what this chapter is all about. However, before we explain that, I will go over some definitions of properties of functions. Firstly, we have *injection*.

Definition 1.2 An injective function, f, or a *one-to-one* function, is a function such that every distinct output of the function corresponds to only *one* distinct input, that is:

$$f(x_1) = f(x_2) \text{ implies } x_1 = x_2$$

By contraposition, $x_1 \neq x_2$ implies $f(x_1) \neq f(x_2)$. The contraposition of a statement "p then q" is "not q then p".

Of course, this explanation might not have been very clear. Here is a visualisation:

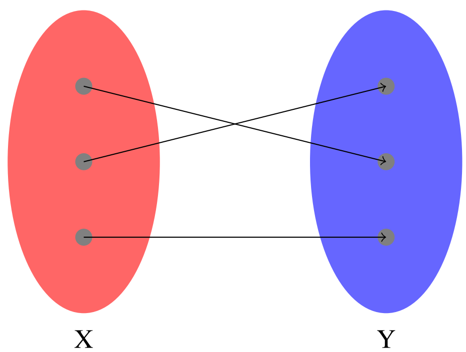


Figure 1.5. Illustration of an injective function with domain X and range Y.

As you can see in Figure 1.5, every output in Y of the function is associated with only one input in X, hence the function is *injective*. (Kind of like how marriages *should* be.) To prove that a function f is injective, you can either show that if $x_1 \neq x_2$, then $f(x_1) \neq f(x_2)$, or if $f(x_1) = f(x_2)$, then $x_1 = x_2$.

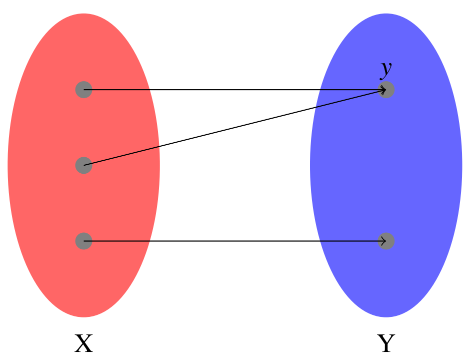


Figure 1.6. Illustration of a non-injective function with domain X and range Y.

Here, the function is *not* injective. As you can see in Figure 1.6, the output y is associated with more than one input in X. With this in mind, it is time to talk about *surjection*.

Definition 1.3 A surjective function f is a function such that for every output y, there exists at least one input x such that $f(x) = y$.

From Figure 1.5 and Figure 1.6, both functions are surjective as every output is associated with at least one input. Do take note that it does not matter how many inputs associate to one output, as long every output is associated with at least one input, the function is surjective. To prove that a function is surjective, you must show that there exists an element in the domain of the function for any arbitrary output in the range.

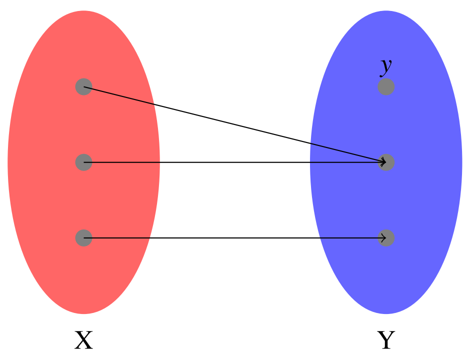


Figure 1.7. Illustration of non-surjective function with domain X and range Y.

From Figure 1.7, we can see that y has no input associated with it. As a result, the function here is not surjective. Lastly, *bijective* functions can't be any easier.

Definition 1.4 A bijective function is a function that is both injective and surjective.

However, bijective functions have a superpower. A function is only bijective *if and only if* it has an inverse.

P iff Q, or "P if and only if Q", means that "if P then Q" and "if Q then P".

Theorem 1.2 A function, $f : A \longrightarrow B$, has an inverse if and only if it is bijection.

PROOF To begin such proofs where there is a "if and only if" clause, we will prove both cases separately. First, we claim that if the function has an inverse, then it is bijective. To prove this, we simply prove that f is injective and surjective (See Lemma 1.1). Then, we claim that if a function is bijective, then it has an inverse (See Lemma 1.2). With both proofs, we show that f has an inverse if and only if it is a bijection. QED

Lemma 1.1 If a function, $f : A \longrightarrow B$, has an inverse, then it is bijective.

PROOF Suppose f^{-1} is the inverse of f. Notice that $f^{-1}(f(x))$ is injective as it is simply $f(x) = x$. Suppose $a, a' \in A$ and $f(a) = f(a')$, then $f^{-1}(f(a)) = f^{-1}(f(a'))$. Notice that by injectivity of $f^{-1}(f(x))$, $a = a'$. Hence, f is injective. Then, suppose $b \in B$, we wish to show that the function is surjective. Also, $f(f^{-1}(x))$ is surjective(it is the identity function). Let $f^{-1}(b) = x$ and $f(f^{-1}(b)) = c$ for some $c \in B$. By a simple substitution, $f(f^{-1}(b)) = f(x) = c$, thereby showing there exists a $x \in A$ for every $c \in B$. The two cases proves that f is bijective if it has an inverse. QED

Lemma 1.2 If a function, $f : A \longrightarrow B$, is a bijection, then it has an inverse.

PROOF Let us define f^{-1} as $f^{-1} : B \longrightarrow A$. Let $b \in B$. Since f is surjective, there exists $a \in A$ such that $f(a) = b$. Let $f^{-1}(b) = a$. Since f is injective, f^{-1} is well-defined, or a function that outputs the same value given the same input. Next, we will show that f^{-1} is indeed the inverse of f by showing the composition of the two is an identity function. Again, let $a \in A$ and $b = f(a)$ which implies, by definition, $f^{-1}(b) = a$. Composing the two functions, $f(f^{-1}(b)) = f(a) = b$, which shows that f^{-1} is indeed the inverse function. The other case of $f^{-1} \circ f$ is similar. Hence, if f is a bijection, then it has an inverse. QED

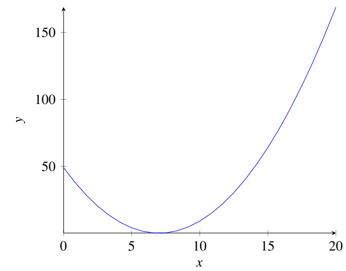


Figure 1.8. Plot of $(x-7)^2$

Remark This serves as a very good example showing how the reader should prove the injectivity, surjectivity, and bijectivity of functions. Also, this explains why in the previous exercise, the function $h(x) = (x-7)^2$ is bounded, only taking x values larger than 7. The function is not bijective in $\mathbb{R}$, as evident in the graph as it is not injective. Hence, the bounds are needed for h to have an inverse.

Exercise

1. Determine if the following functions are injective or surjective or bijective in its domain:
   (a) $f(x) = \sqrt{x}, f : \mathbb{R}^+ \to \mathbb{R}^+$
   (b) $f(x) = \frac{1}{x}, x > 0$
   (c) $f(x) = (x-7)^2, x \geq 5$
2. *Prove* that:
   (a) The composition of two injective functions is *injective*.
   (b) The inverse of a function is *injective*.
   (c) The inverse of a function is *surjective*.

Solution

1. (a) Bijective
   (b) Injective
   (c) Surjective
2. (a) Let the two functions be $f : A \to B$ and $g : B \to C$. If $g(f(x_1)) = g(f(x_2))$, by injectivity of g, $f(x_1) = f(x_2)$ and by injectivity of f, $x_1 = x_2$. Thus, $g \circ f$ is injective.
   (b) Let $f : A \to B$ and f^{-1} be its inverse. Notice if $f^{-1}(x_1) = f^{-1}(x_2)$, then $f(f^{-1}(x_1)) = f(f^{-1}(x_2))$ which implies $x_1 = x_2$. Hence, f^{-1} is injective.
   (c) Similarly, define f and f^{-1}. Let $y \in A$ and $f^{-1}(x) = y$. Notice that $x = f(y)$ and because f is surjective(by Lemma 1.1), $x \in B$, thus proving that f^{-1} is surjective.

And this concludes the chapter on functions. Readers are encouraged to do more similar practices to strengthen their knowledge on functions. Also, this chapter is more on the "pure mathematics" side, such proofs are only presented as a good exercise to the reader. In actuality, the book does not focus that much on this kind of mathematics(i.e. analysis) but more on the practical aspects of Calculus.

SECTION 1.4
An aside: A more rigorous definition of a function

TBD

CHAPTER 2
Linear Algebra

SECTION 2.1
Vector Basics

Firstly, good job on finishing chapter 1, it is not an easy feat. Stepping into this chapter, you may feel overwhelmed by the name, *Linear Algebra*. What is that? Please take note though, the knowledge in this chapter will not be used until Calculus III to IV in part IV of the book. If you are just willing to learn Calc I and II, you do not need the knowledge here. With that said, let's delve into Vectors. A concept that is very ubiquitous in this world is a coordinate, like the GPS coordinate in the maps app. A coordinate simply describes the position of a point in space. Generally, it is denoted (x, y) in 2 dimensions and (x, y, z) in 3 dimensions. Likewise, vectors are also behaves like coordinates and are written down in a *similar* fashion. Think of a vector as a coordinate.

Definition 2.1 A vector is a mathematical object with a magnitude, or length, and a direction. A vector, $\vec{a}$ is a point (x, y) in space drawn with an arrow. It is denoted as:

$$\vec{a} = \begin{bmatrix} x \\ y \end{bmatrix}$$

Likewise, a vector at (x, y, z) is denoted $\begin{bmatrix} x \\ y \\ z \end{bmatrix}$

Of course, vectors by themselves are not very useful. Hence, we shall define the vector space. For the purposes of this text, you may think of vector spaces as the 2 and 3 dimensional euclidean spaces you are very familiar with. In a vector space, we can finally meaningfully manipulate them. Take a look at the vector $\vec{a} = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$ and the vector $\vec{b} = \begin{bmatrix} -2 \\ 1 \end{bmatrix}$ in the two dimensional euclidean space:

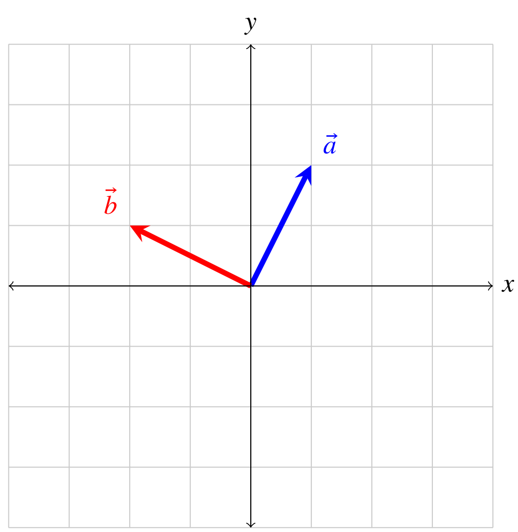


Figure 2.1. Figure of vectors $\vec{a}$ and $\vec{b}$.

You may have noticed that the coordinate axes in Figure 2.1 are the same as the ones you are familiar with. However, in linear algebra, instead of the usual xy axes to describe the components of a vector, you have $\hat{\mathbf{i}}$ and $\hat{\mathbf{j}}$ to describe the components of a vector.(So instead of saying the "x" component of a vector, you would say the "i hat" component of a vector.) This is a convention in linear algebra. $\hat{\mathbf{\imath}}$ translates to the x axis and $\hat{\mathbf{\jmath}}$ translates to the y axis. Likewise, in the third dimension, $\hat{\mathbf{k}}$ represents the z axis. Another thing to note are the *basis vectors*:

Some authors use $\hat{x}, \hat{y}, \hat{z}$ instead of the usual $\hat{\imath}, \hat{\jmath}, \hat{k}$. We shall use the former.

Definition 2.2 The set of vectors in a vector space are called *basis* if every vector in the vector space can be written uniquely written as a finite *linear combination* of the basis vectors. An example would be these vectors $\hat{\mathbf{\imath}} = \left[\begin{smallmatrix}0\\1\end{smallmatrix}\right]$ and $\hat{\mathbf{\jmath}} = \left[\begin{smallmatrix}1\\0\end{smallmatrix}\right]$ in the 2 dimensional euclidean space which can be thought of one unit of distance in the x and y direction respectively.

A linear combination of x and y is $ax+by$ where a, b are constants.

With this in mind, another way of writing a vector $\vec{a} = \left[\begin{smallmatrix}x\\y\end{smallmatrix}\right]$ is $\vec{a} = x\hat{\mathbf{\imath}} + y\hat{\mathbf{\jmath}}$. Notice that like in Definition 2.2, $\hat{\mathbf{\imath}}$ and $\hat{\mathbf{\jmath}}$ are written in a linear combination of x and y. Of course, these are not the only basis vectors in 2d, if you have learned polar coordinates before, there is the equivalent $\hat{\mathbf{r}}$ and $\hat{\boldsymbol{\theta}}$, but this is a topic for a later chapter. Next, we shall look at the *addition* of vectors. A vector $\vec{a} = \left[\begin{smallmatrix}x_a\\y_a\end{smallmatrix}\right]$ added to a vector $\vec{b} = \left[\begin{smallmatrix}x_b\\y_b\end{smallmatrix}\right]$ is the vector $\left[\begin{smallmatrix}x_a+x_b\\y_a+y_b\end{smallmatrix}\right]$. As you can see, the addition of two vectors results in another vector. Here is a illustration to help clear up things:

You can think of $\hat{\imath}$ and $\hat{\jmath}$ as one unit of direction in the x and y axis respectively.

This can be extrapolated into any dimensions: $\left[\begin{smallmatrix}a\\b\\c\end{smallmatrix}\right] + \left[\begin{smallmatrix}d\\e\\f\end{smallmatrix}\right] = \left[\begin{smallmatrix}a+d\\b+e\\c+f\end{smallmatrix}\right]$.

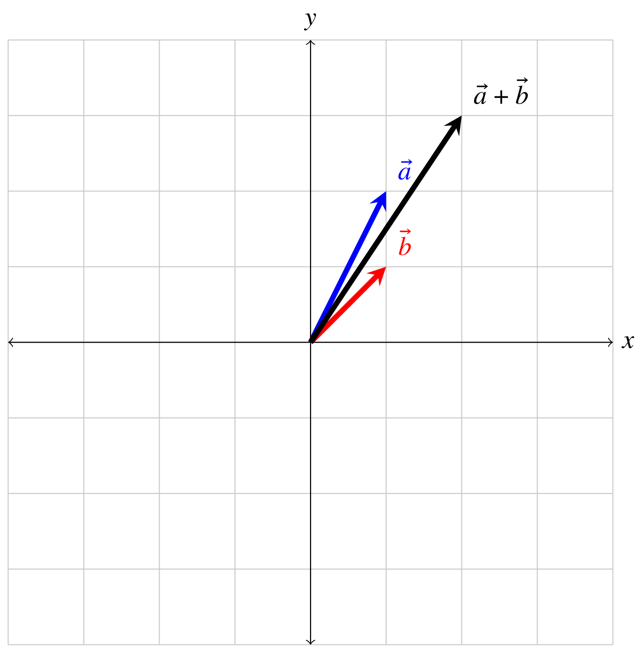


Figure 2.2. Illustration of vector addition.

Let us analyze Figure 2.2. Firstly, notice that $\vec{a} = 1\hat{\mathbf{\imath}}+2\hat{\mathbf{\jmath}}$ and $\vec{b} = 1\hat{\mathbf{\imath}}+1\hat{\mathbf{\jmath}}$. Thus, $\vec{a}+\vec{b} = 2\hat{\mathbf{\imath}}+3\hat{\mathbf{\jmath}}$, exactly like in our figure. One way to understand this is to "attach" the vector $\vec{a}$'s tail to $\vec{b}$'s head by translating $\vec{a}$. You may have seen this in your physics class before. Similarly, the subtraction of vectors $\vec{x} = \left[\begin{smallmatrix}a\\b\end{smallmatrix}\right]$ and $\vec{y} = \left[\begin{smallmatrix}c\\d\end{smallmatrix}\right]$ is $\vec{x} - \vec{y} = \left[\begin{smallmatrix}a-c\\b-d\end{smallmatrix}\right]$. Of course, we can use the same "head to tail" model to explain vector subtraction. Subtracting $\vec{y}$ from $\vec{x}$ is simply finding the vector whose tail is on the head of $\vec{y}$ and whose head is on the head of $\vec{x}$. However, some of you may be scratching your heads right now, what if a vector does not start from the *origin* (or the coordinate $(0,0)$) but its tail lands on, say $(1,1)$. How do we express such a vector? The answer may be surprising, both vectors have the same representation. However, unless specifically stated, *all* vectors start from the origin. This is also why the "head to tail" model works, while the "tail" of the vector is not at the origin, translating it to the origin results in an *identical* vector.

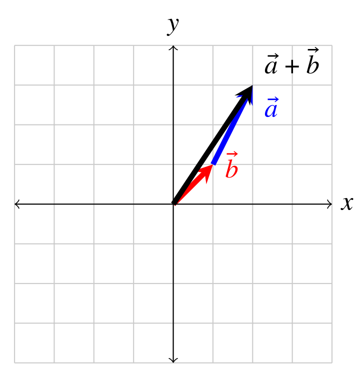


Figure 2.3. Illustration of vector "head to tail" addition.

Translation means moving something without rotation or any scaling.

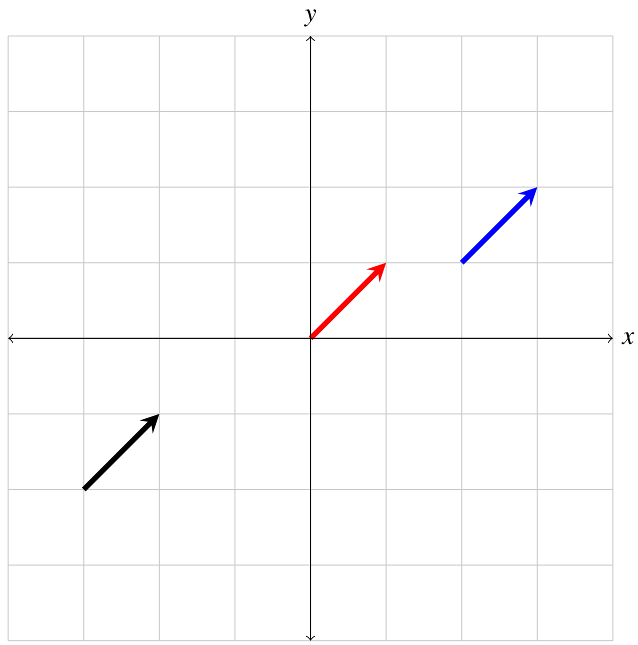


Figure 2.4. Three identical vectors.

Example Here are some examples of vectors:

$$\vec{a} = \begin{bmatrix} 1 \\ 2 \end{bmatrix} \ \vec{b} = \begin{bmatrix} -3 \\ 4 \end{bmatrix} \ \vec{c} = \begin{bmatrix} 2 \\ -2 \end{bmatrix}$$

Here are some examples of adding and subtracting vectors. Note that these operations are *commutative*, in other words the order does not matter:

$$\vec{a} + \vec{b} = \begin{bmatrix} 1-3 \\ 2+4 \end{bmatrix} = \begin{bmatrix} -2 \\ 6 \end{bmatrix}$$

$$\vec{a} - \vec{c} = \begin{bmatrix} -1 \\ 4 \end{bmatrix}$$

$$\vec{a} + \vec{b} - \vec{c} = \begin{bmatrix} -2 \\ 6 \end{bmatrix} - \vec{c} = \begin{bmatrix} -4 \\ 8 \end{bmatrix}$$

Given a vector $\vec{a}$, $-\vec{a}$ can be thought of the same vector with the same magnitude but pointing in the *opposite* direction. To find it, simply flip the signs of all the components in $\vec{a}$. If you add $\vec{a}$ to $-\vec{a}$, you get what is called the *zero vector*, or the vector $\left[\begin{smallmatrix} 0 \\ 0 \end{smallmatrix}\right]$. This is a vector without direction has zero magnitude. We shall denote it $\vec{0}$. Lastly, I want to briefly touch on the idea of *magnitude* which is basically the length of a vector. The derivation of the following formula is simply the Pythagorean theorem.

Definition 2.3 The magnitude of a vector $\vec{v}$ in n-dimensional space is as such:

$$||\vec{v}|| = \sqrt{v_1^2 + v_2^2 + \ldots + v_n^2}$$

A *unit* vector is a vector with magnitude 0.

Try to attempt the following exercise:

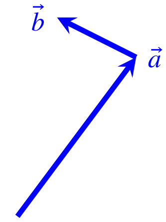


Figure 2.5. Visualisation of "head to tail" addition of $\vec{a}$ and $\vec{b}$.

Exercise

1. Given that $\vec{a} = \begin{bmatrix} 3 \\ 4 \end{bmatrix}$ and $\vec{b} = \begin{bmatrix} -2 \\ 1 \end{bmatrix}$ and $\vec{c} = \begin{bmatrix} -3 \\ -2 \end{bmatrix}$, calculate:
   (a) $\vec{a} + \vec{b}$
   (b) $\vec{a} - \vec{c}$
   (c) $\vec{a} - \vec{b} - \vec{c}$
2. Are the following statements true or false:
   (a) The subtraction of two vectors *always* results in a vector.
   (b) When any vector is added to the zero vector, the result *always* is not a zero vector.
3. Find the magnitude of $\begin{bmatrix} 3 \\ 4 \end{bmatrix}$ and $\begin{bmatrix} 3 \\ 4 \\ 5 \end{bmatrix}$.
4. **(Difficult)** Prove that the addition of vectors is *associative* given that the addition in $\mathbb{R}$ is associative.

An operation $\circ$ is associative if order in which it is applied does not matter, in this case $(\vec{x} + \vec{y}) + \vec{z} = \vec{x} + (\vec{y} + \vec{z})$.

Solution

1. (a) $\begin{bmatrix} 1 \\ 5 \end{bmatrix}$
   (b) $\begin{bmatrix} 6 \\ 6 \end{bmatrix}$
   (c) $\begin{bmatrix} 8 \\ 7 \end{bmatrix}$
2. (a) True since subtraction can be viewed as $\vec{a} + (-\vec{b})$ and $-\vec{b}$ is a vector.
   (b) False since $\vec{0} + \vec{0} = \vec{0}$.
3. 5 and $\sqrt{50}$.
4. Let $\vec{x}, \vec{y}, \vec{z}$ be vectors in $\mathbb{R}^n$ (N-dimensional euclidean space). Notice that:

$$
\begin{aligned}
(\vec{x} + \vec{y}) + \vec{z} &= \left(\begin{bmatrix} x_1 \\ \dots \\ x_n \end{bmatrix} + \begin{bmatrix} y_1 \\ \dots \\ y_n \end{bmatrix} \right) + \begin{bmatrix} z_1 \\ \dots \\ z_n \end{bmatrix} \\
&= \begin{bmatrix} x_1 + y_1 + z_1 \\ \dots \\ x_n + y_n + z_n \end{bmatrix} \\
&= \begin{bmatrix} x_1 + (y_1 + z_1) \\ \dots \\ x_n + (y_n + z_n) \end{bmatrix} \\
&= \begin{bmatrix} x_1 \\ \dots \\ x_n \end{bmatrix} + \left(\begin{bmatrix} y_1 \\ \dots \\ y_n \end{bmatrix} + \begin{bmatrix} z_1 \\ \dots \\ z_n \end{bmatrix} \right) \\
&= \vec{x} + (\vec{y} + \vec{z})
\end{aligned}
$$

Section 2.2
Scaling and dot product

In this chapter we will deal two kinds of multiplication with vectors. By far the easiest would be scaling a vector. As we have previously discussed, the magnitude of a vector is the length of a vector. But what if I want to "elongate" the vector by some amount to make it longer or shorter. This is called *scaling*.

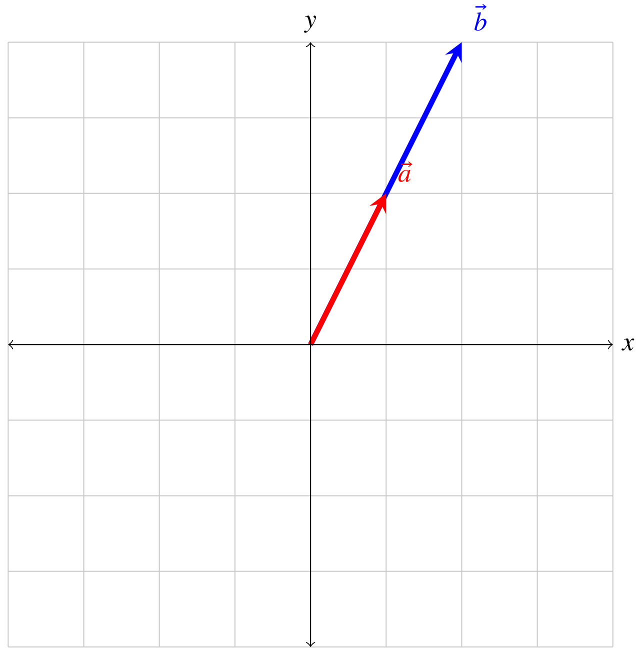


Figure 2.6. Two vectors, one twice as long as the other.

Take a look at Figure 2.6, $\vec{b}$ is twice as long as $\vec{a}$. What should we multiply to $\vec{a}$ to make as long as $\vec{b}$? If you guessed 2, then you are right. (Not sure what else you could have guessed.) To multiply a vector by a constant, we multiply each individual component of the vector by the constant. From the figure, we can identify that $\vec{a} = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$ so $2\vec{a} = \begin{bmatrix} 2\times 1 \\ 2\times 2 \end{bmatrix} = \begin{bmatrix} 2 \\ 4 \end{bmatrix}$ which is equal to $\vec{b}$. You may extrapolate this to higher dimensions.

Definition 2.4 Scaling a n-dimensional vector $\vec{v}$ by a constant n is defined as such:

$$n\vec{v} = n\begin{bmatrix} v_1 \\ \cdots \\ v_n \end{bmatrix} = \begin{bmatrix} n\times v_1 \\ \cdots \\ n\times v_n \end{bmatrix}$$

Never ever write $n \times \vec{v}$ or $n\dot{\vec{v}}$ or you will be cursed into the thundering depths of careless *misteaks* everywhere.

Let us try something fun, what if we multiplied the vector by a negative number? Well, the computation remains the same. As you can see from Figure 2.7, $\vec{a} = (-0.5)\vec{b}$. You may wish to compute this yourself to verify. One thing to note here is that the vector gets flipped across the origin when the constant the vector is multiplied to is negative. Also, when scaling a vector by some constant, the reader may want to verify using Definition 2.3 that the magnitude of the vector is also scaled by that constant. This should be a trivial proof.

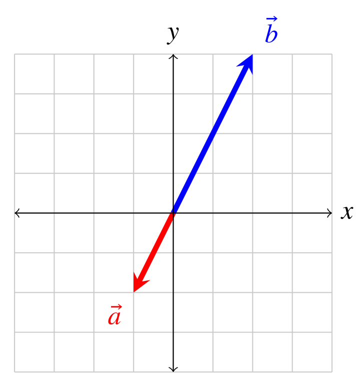


Figure 2.7. Two vectors, one is scaled by -0.5.

You may have noticed that I have abstained from using either the $\cdot$ or $\times$ operators. The reason is bad notation which is sprinkled throughout mathematics. These two operators have special meanings in the realm of linear algebra. We shall first explore the easier one, the *dot product*.

Dot products use the $\cdot$ operator and only acts on vectors, do not abuse notation by using the $\cdot$ operator in the context of scaling a vector. Dot products are actually fairly straightforward to compute:

Definition 2.5 The dot product of two vectors in n-dimensional space $\vec{a}$ and $\vec{b}$ is:

$$\vec{a} \cdot \vec{b} = a_1 b_1 + a_2 b_2 + \cdots + a_n b_n = (||\vec{a}||)(||\vec{b}||)\cos\theta$$

where θ is the angle between the two vectors. We shall refer to the formula between the first and second equal signs the "first definition of dot products" and the formula after the second equal sign the "second definition of dot products".

From Definition 2.5, we can tell that the dot product tells us something about the relationship between the two vectors. In essence, if the dot product between two vectors is large and positive, then the vectors are pointing in roughly the same direction. Else, if the product is very negative, then the dot product is very negative.

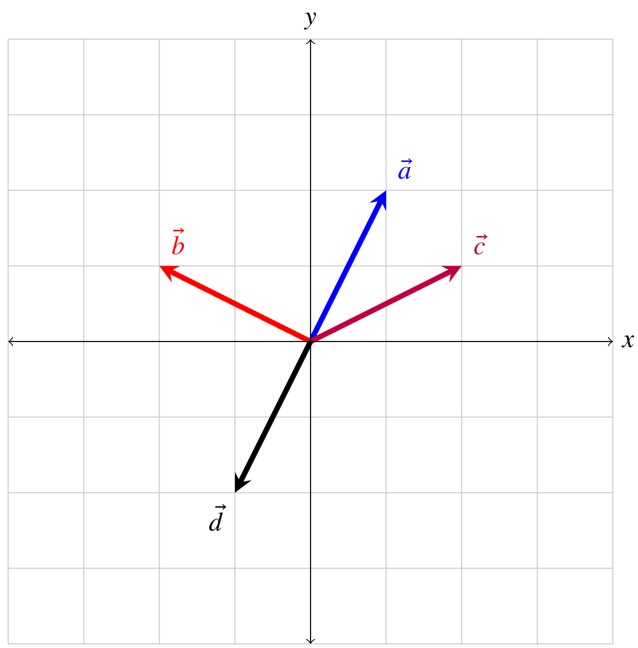


Figure 2.8. Four vectors pointing in different directions.

Firstly, note down the components of each vector in Figure 2.8. Then, let us compute $\vec{a} \cdot \vec{b}$. By Definition 2.5, it is $1(-2) + 2(1) = 0$. Also, it is important to take note that these two vectors are orthogonal(at right angles with each other). This brings us to an important property of dot products, if two vectors are orthogonal, then their dot product is 0. This can be seen from the fact that $\cos\left(\frac{\pi}{2}\right) = 0$ and from the second definition of dot products in Definition 2.5. In a sense, the dot product tells you how aligned are the directions of two vectors. Another example would be taking the dot product of two identical vectors, such as $\vec{a} \cdot \vec{a}$. The dot product there is $1(1) + 2(2) = 5$. In addition, we know $\cos 0 = 1$ so according to the second definition, $\vec{a} \cdot \vec{a} = (||\vec{a}||)(||\vec{a}||) = (||\vec{a}||)^2$. If we take the square root on both sides, $||\vec{a}|| = \sqrt{\vec{a} \cdot \vec{a}}$ which can be rewritten as $||\vec{a}|| = \sqrt{a_1^2 + \cdots + a_n^n}$. Drum roll please. This is exactly the formula we got in Definition 2.3 for the magnitude of a vector. This is a pretty nice result.

Example

From Figure 2.8, we can see that:

$$\vec{a} \cdot \vec{c} = \begin{bmatrix} 1 \\ 2 \end{bmatrix} \cdot \begin{bmatrix} 2 \\ 1 \end{bmatrix} = 1(2) + 2(1) = 4$$

$$\vec{a} \cdot \vec{d} = \begin{bmatrix} 1 \\ 2 \end{bmatrix} \cdot \begin{bmatrix} -1 \\ -2 \end{bmatrix} = 1(-1) + 2(-2) = -5$$

Remark Generally, when finding the dot product, we use the first form in Definition 2.5 as it is typically easier to compute. It is also important to know that the dot product of two vectors results in a number, not a vector like in scaling.

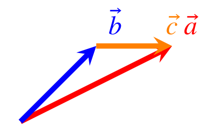


Figure 2.9. Figure of a triangle formed by three vectors.

Exercise

1. If $\vec{x} = \left[\begin{smallmatrix} 1 \\ 2 \end{smallmatrix}\right]$ and $\vec{b} = \left[\begin{smallmatrix} 3 \\ 4 \end{smallmatrix}\right]$, compute the following:
   (a) $2\vec{x}$
   (b) $\vec{x} \cdot \vec{y}$
   (c) $\vec{y} \cdot \vec{x}$
2. Prove or disprove that the dot product *commutes*.
3. Given two vectors $\vec{a}$ and $\vec{b}$ separated by an angle θ, they form a triangle with a third side $\vec{c} = \vec{a} - \vec{b}$ (See Figure 2.9).By taking the dot product of $\vec{c}$ and itself, derive the law of cosines. (Hint: $(\vec{x} \cdot \vec{y}) \cdot (\vec{z} \cdot \vec{w}) = \vec{x} \cdot \vec{z} + \vec{x} \cdot \vec{w} + \vec{y} \cdot \vec{z} + \vec{y} \cdot \vec{w}$)

An operation $\circ$ is commutative if $x \circ y = y \circ x$.

Solution

1. (a) $\begin{bmatrix} 2 \\ 4 \end{bmatrix}$
   (b) 11
   (c) 11
2. The dot product commutes. From the second definition of the dot product from Definition 2.5, we can see that $\vec{a} \cdot \vec{b} = (||\vec{a}||)(||\vec{b}||)\cos(\theta) = (||\vec{b}||)(||\vec{a}||)\cos(\theta) = \vec{b} \cdot \vec{a}$. Therefore, the dot product commutes.
3. Let a, b, c be the magnitudes of the respective vectors. Just take the dot product of $\vec{c}$ and itself.

$$\begin{aligned} \vec{c} \cdot \vec{c} &= \left(\vec{a} - \vec{b}\right) \cdot \left(\vec{a} - \vec{b}\right) \\ &= a^2 - 2\vec{a} \cdot \vec{b} + b^2 \\ &= a^2 + b^2 - 2ab\cos(\theta) \end{aligned}$$

Which is exactly the law of cosines.

Note that $\vec{a} \cdot \vec{a} = (||\vec{a}||)^2$ as evident from the second definition of Definition 2.5.

SECTION 2.3
Matrices

In this chapter, we will introduce a new mathematical object that you have probably heard before but not sure what it is. To be honest, many mathematics students only know how to *compute* matrix operations but does not know the meaning behind them. I will try to do my best to explain them in a clear manner and I hope that you would appreciate matrices more for their beautiful geometric meaning than their mind-numbing calculations. A great place to start is the *basis vectors* introduced back in Definition 2.2. All vectors can be written as a unique linear combination of the basis vectors. But $\hat{\mathbf{i}}$ and $\hat{\mathbf{j}}$ are not the only basis vectors out there, what if I wanted to describe vectors using another set of basis vectors? Perhaps that would be more convenient for my special use case. Well, it is actually pretty easy to do so. For the sake of this argument, let us try to rotate all of the vectors in a normal 2d vector field anti-clockwise by 90°. Notice how we could not describe $\vec{b}$ as some constant multiplied to $\vec{a}$ in Figure 2.10. We need some other tool to help us. If we change the basis vectors of $\vec{a}$ to another pair of basis vectors, can we somehow get $\vec{b}$? Here is a thought: replace the $\hat{\mathbf{i}}$ with $\left[\begin{smallmatrix}0\\1\end{smallmatrix}\right]$ and $\hat{\mathbf{j}}$ with $\left[\begin{smallmatrix}-1\\0\end{smallmatrix}\right]$. So, $\vec{a} = 1\hat{\mathbf{i}}+2\hat{\mathbf{j}}$ will become $\vec{a} = \left[\begin{smallmatrix}0\\1\end{smallmatrix}\right] + 2\left[\begin{smallmatrix}-1\\0\end{smallmatrix}\right] = \left[\begin{smallmatrix}-2\\1\end{smallmatrix}\right]$. This is pretty intriguing, by changing the basis vectors, we have basically "morphed" $\vec{a}$ into $\vec{b}$. This "morphing" is called a *linear transformation* or *linear map*. In some of the visualisations of the linear transformations I might keep the original grid in the background in a lighter color.

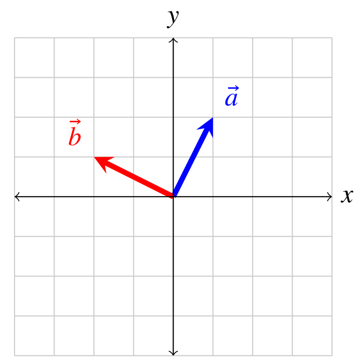


Figure 2.10. Two vectors, one rotated anti-clockwise by 90°.

Example

Here are some examples of linear transformations, the blue grid is the vector space after the transformation. Take note where the basis vectors are after the transformations.

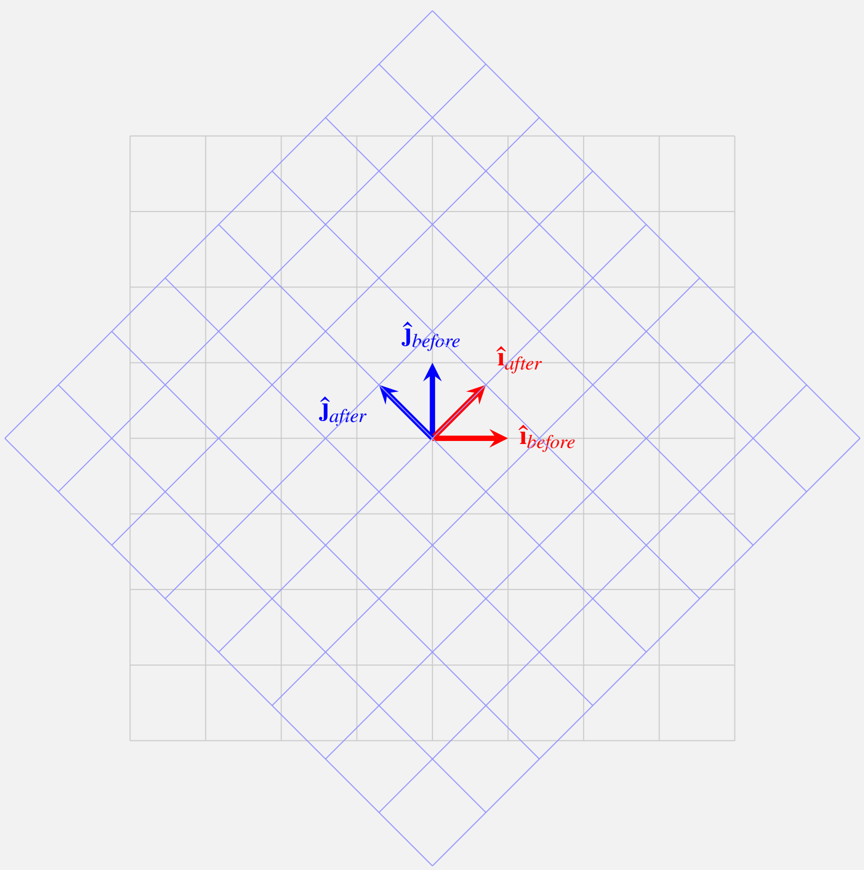


Figure 2.11. A transformation in the form of an anti-clockwise *rotation*.

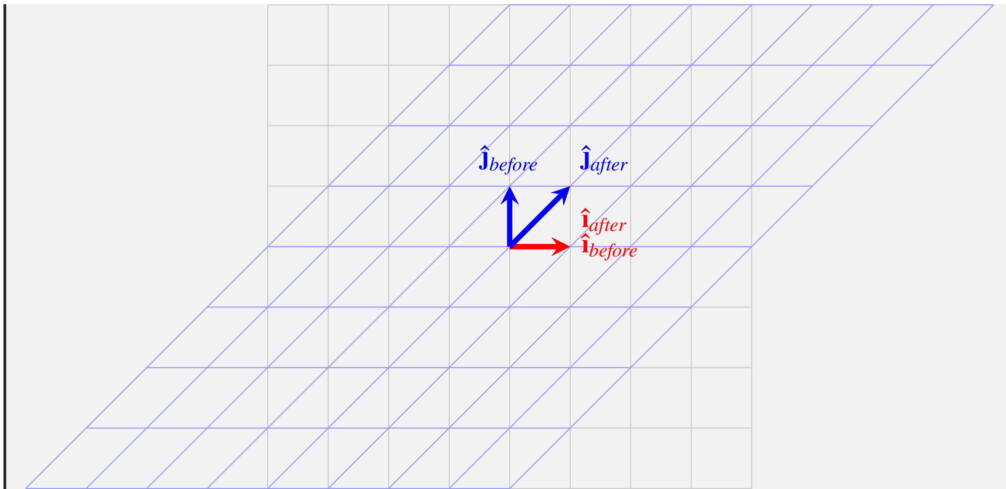


Figure 2.12. A transformation in the form of a *shear*.

In Figure 2.12, the $\hat{\imath}$ vector remains at the same location.

To describe these transformations, we actually only need to describe where the basis vectors land after the transformation. No other information required. We present the location of the new basis vectors in the form of a *matrix*:

Definition 2.6 In linear algebra, matrices generally represent linear transformations. A $n \times m$ matrix is one with n rows and m columns. The nth column of the matrix denotes the representation of the corresponding basis vector after the linear transformation.

$$\begin{bmatrix} a & b \\ c & d \end{bmatrix}$$

represents a matrix that moves the $\hat{\imath}$ basis vector to $\left[\begin{smallmatrix} a \\ c \end{smallmatrix}\right]$ and moves the $\hat{\jmath}$ basis vector to $\left[\begin{smallmatrix} b \\ d \end{smallmatrix}\right]$.

Remark Matrices can also represent a variety of things apart from linear transformations. It is just a mathematical object representing a bunch of things arranged in rows and columns.

It should go without saying that any vector in the original vector space will get moved after a transformation. To calculate where a vector ends up after a transformation, we multiply the matrix by the vector. This is where many textbooks make linear algebra feel laborious. In order to make the process more intuitive, recall that a vector has a $\hat{\imath}$ and $\hat{\jmath}$ component. Now, we swap the basis vectors for the new basis vectors described in our matrix. This is just multiplying a vector by a constant, which you have done before.

Example Take the transformation in Figure 2.12 which is represented by the matrix $\left[\begin{smallmatrix} 1 & 1 \\ 0 & 1 \end{smallmatrix}\right]$. Let us say in the original vector space, there is a vector $\vec{a} = \left[\begin{smallmatrix} 1 \\ 2 \end{smallmatrix}\right]$. Then, the vector's representation after the transformation:

$$\begin{aligned} \begin{bmatrix} 1 \\ 2 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix} &= 1 \begin{bmatrix} 1 \\ 0 \end{bmatrix} + 2 \begin{bmatrix} 1 \\ 1 \end{bmatrix} \\ &= \begin{bmatrix} 3 \\ 2 \end{bmatrix} \end{aligned}$$

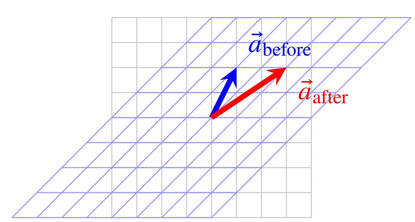


Figure 2.13. Figure of a vector before and after a shear transformation like in Figure 2.12.

If you have come from a linear algebra course, this may feel very refreshing. Instead of memorising what needs to be multiplied, you just swap the basis vectors for some other basis vectors. While we get the representation of $\vec{a}$ from Figure 2.13, it also important to take note that in the new vector space after the transformation, $\vec{a}$ has one unit of direction in the horizontal direction and two units of direction in the "vertical" direction. This is just like the original representation of $\vec{a}$, just that the vector space has been sheared. This change of basis vector is important enough to have its own name, *change of basis*.

With this knowledge, let us proceed to one of the most hated parts of linear algebra, *matrix multiplication*. These few words can bring out so much frustration out of so many students. It is quite a laborious and boring process if you do not grasp the meaning and motivation behind it. In actuality, the idea is simple. When you multiply two matrices together, you are effectively applying two linear transformations, one after another. The result is also a matrix, which is a linear transformation, hence the operation is *closed*.

This is only true if the matrix is $n \times n$, which are the ones we are dealing with.

Example Say you have two matrices, a 90°anti-clockwise rotation $x = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$ and a horizontal shear $y = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$. xy is simply applying the shear first before the rotation. (The right to left reading is similar to that of function composition)

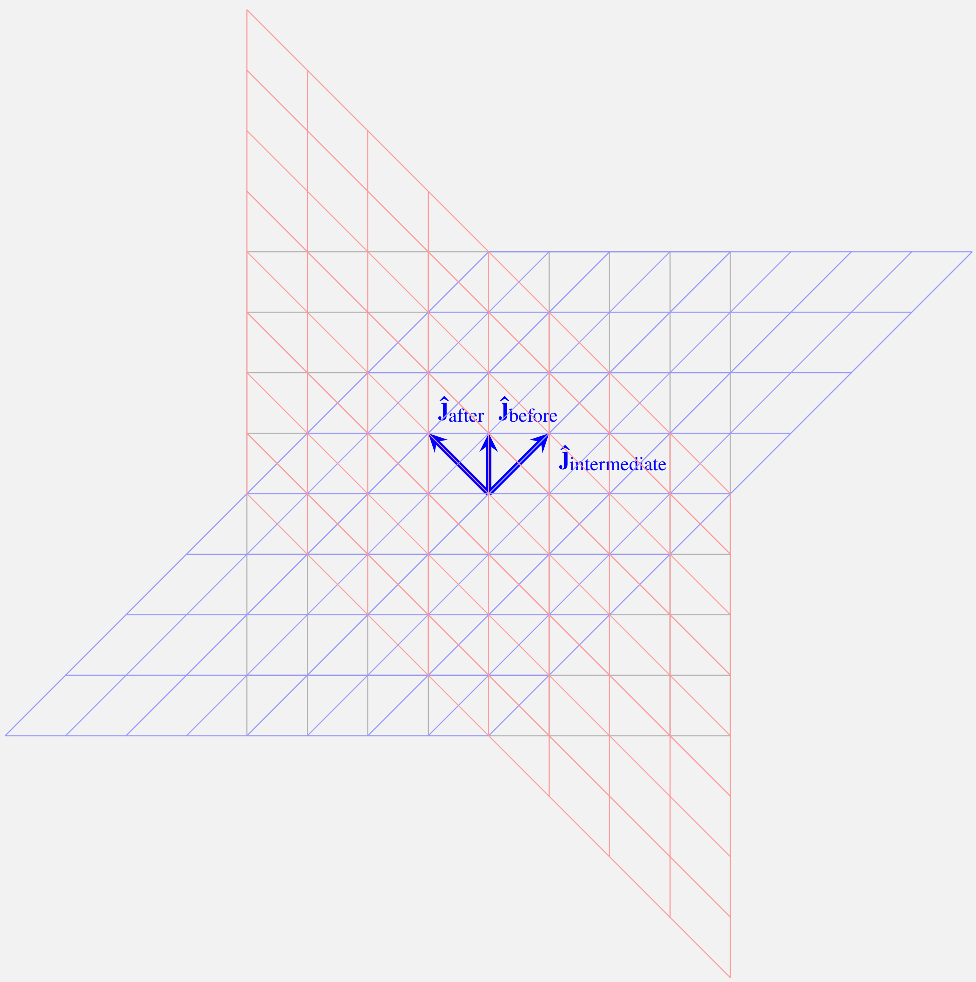


Figure 2.14. Figure of applying two linear transformations.

It should not be difficult to see that the end result of applying the two linear transformations in the previous examples resulted in another linear transformation (the red grid). We wish to find the representation of this linear transformation as a matrix. Let us go through this step by step. First of all, the basis vectors of the *identity matrix* , gets moved to some new location. In other words, given a matrix $a = \begin{bmatrix} a_1 & a_2 \\ a_3 & a_4 \end{bmatrix}$, we know that the basis vectors gets moved to $\begin{bmatrix} a_1 \\ a_3 \end{bmatrix}$ and $\begin{bmatrix} a_2 \\ a_4 \end{bmatrix}$ respectively for the $\hat{\mathbf{i}}$ and $\hat{\mathbf{j}}$ vectors. Now, we have another transformation $b = \begin{bmatrix} b_1 & b_2 \\ b_3 & b_4 \end{bmatrix}$. To know where the $\hat{\mathbf{i}}$ vector lands after both transformations, notice that it is simply multiplying the

The identity matrix is simply the transformation that leaves the vector space unchanged.

intermediate position of it by b, which is $\begin{bmatrix} b_1 & b_2 \\ b_3 & b_4 \end{bmatrix}\begin{bmatrix} a_1 \\ a_3 \end{bmatrix} = \begin{bmatrix} a_1b_1+a_3b_2 \\ a_1b_3+a_3b_4 \end{bmatrix}$. A similar line of reasoning may be used for $\hat{\mathbf{j}}$. Hence, the result is:

$$ba = \begin{bmatrix} a_1b_1 + a_3b_2 & a_2b_1 + a_4b_2 \\ a_1b_3 + a_3b_4 & a_2b_3 + a_4b_4 \end{bmatrix}$$

This is the reason many students find linear algebra tedious. Most courses make students memorise the formula which is extremely rote. I wish that you understand what is going on behind the scenes on this formula. With that said, it is best to get some experience with these multiplications:

Example

Given that $x = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$ and $y = \begin{bmatrix} 1 & 3 \\ 2 & 4 \end{bmatrix}$ and $z = \begin{bmatrix} 5 & 6 \\ 7 & 8 \end{bmatrix}$,

$$xy = \begin{bmatrix} 5 & 11 \\ 11 & 25 \end{bmatrix}$$
$$yx = \begin{bmatrix} 10 & 14 \\ 14 & 20 \end{bmatrix}$$
$$x(yz) = \begin{bmatrix} 102 & 118 \\ 230 & 266 \end{bmatrix}$$
$$(xy)z = \begin{bmatrix} 102 & 118 \\ 230 & 266 \end{bmatrix}$$

This examples shows that matrix multiplication does not commute but it is associative.

Exercise

1. If $\vec{x} = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$, $a = \begin{bmatrix} 2 & 3 \\ 4 & 5 \end{bmatrix}$ and $b = \begin{bmatrix} 6 & 7 \\ 8 & 9 \end{bmatrix}$, compute:
   (a) $a\vec{x}$
   (b) ab
   (c) $ab\vec{x}$
2. **(Difficult)** Show that matrix multiplication is *associative* for all 2×2 matrices.

Solution

1. (a) $\begin{bmatrix} 8 \\ 14 \end{bmatrix}$
   (b) $\begin{bmatrix} 36 & 41 \\ 64 & 37 \end{bmatrix}$
   (c) $\begin{bmatrix} 118 \\ 210 \end{bmatrix}$
2. Let $x = \begin{bmatrix} x_1 & x_2 \\ x_3 & x_4 \end{bmatrix}$ and define y and z likewise, notice that:

$$\begin{aligned} x(yz) &= \begin{bmatrix} x_1 & x_2 \\ x_3 & x_4 \end{bmatrix}\begin{bmatrix} z_1y_1 + z_3y_2 & z_2y_1 + z_4y_2 \\ z_1y_3 + z_3y_4 & z_2y_3 + z_4y_4 \end{bmatrix} \\ &= \begin{bmatrix} x_1 & x_2 \\ x_3 & x_4 \end{bmatrix}\begin{bmatrix} a_1 & a_2 \\ a_3 & a_4 \end{bmatrix} \quad \text{let a=}yz \\ &= \begin{bmatrix} a_1x_1 + a_3x_2 & a_2x_1 + a_4x_2 \\ a_1x_3 + a_3x_4 & a_2x_3 + a_4x_4 \end{bmatrix} \\ &= (xy)z \end{aligned}$$

The derivation for $(xy)z$ will be left as an exercise for the reader. Both should evaluate to the same expression.

Section 2.4

Determinants, inverses and Cramer's rule

At this juncture, you should be familiar with the concept that a matrix represents a linear transformation and it morphs the vector space. Also, if you imagine the basis vectors of any vector space as lengths of a parallelogram, then you might want to find the area of that parallelogram. For example, the identity matrix's basis vectors forms a square with area 1. Likewise, given a matrix $x = \left[\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right]$, we can display it as:

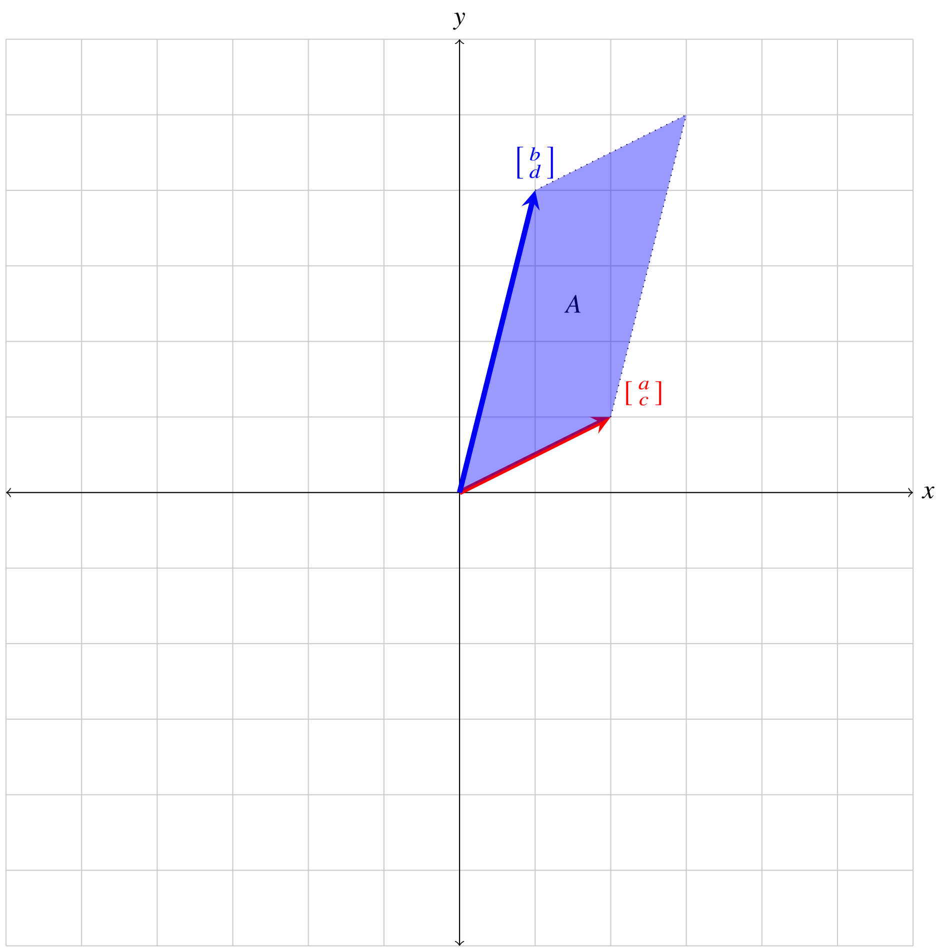


Figure 2.15. Figure of area A formed by a pair of basis vectors.

To find the area of A in Figure 2.15, we may use the sine rule (let $\vec{u} = \left[\begin{smallmatrix} a \\ c \end{smallmatrix}\right]$ and $\vec{v} = \left[\begin{smallmatrix} b \\ d \end{smallmatrix}\right]$):

$$\begin{aligned}
A &= (||\vec{u}||)(||\vec{v}||)\sin(\theta) \quad \text{where } \theta \text{ is the angle between the vectors} \\
&= (||\vec{u'}||)(||\vec{v}||)\cos(\theta') \quad \text{where } \vec{u'} \text{ is } \left[\begin{smallmatrix} a \\ -c \end{smallmatrix}\right]\text{, perpendicular to } \vec{v} \text{ and } \theta' \text{ is } 90^\circ \\
&= \begin{bmatrix} -c \\ a \end{bmatrix} \begin{bmatrix} b \\ d \end{bmatrix} \quad \text{by Definition 2.5, definition of dot product} \\
&= ad - bc
\end{aligned} \tag{2.1}$$

The resultant area is also known as the *determinant* of a matrix.

Definition 2.7 The determinant of a 2×2 matrix $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$ is represented as:

$$\det(A) = \begin{vmatrix} a & b \\ c & d \end{vmatrix} = ad - bc$$

and it represents the signed area of a parallelogram with sides of the basis vectors described in A. Meanwhile, a 3×3 matrix $B = \begin{bmatrix} a & b & c \\ d & e & f \\ g & h & i \end{bmatrix}$ has a determinant of:

$$\begin{aligned} \det(B) &= \begin{vmatrix} a & b & c \\ d & e & f \\ g & h & i \end{vmatrix} \\ &= a\begin{vmatrix} e & f \\ h & i \end{vmatrix} - b\begin{vmatrix} d & f \\ g & i \end{vmatrix} + c\begin{vmatrix} d & e \\ g & h \end{vmatrix} \end{aligned}$$

and it represents the volume of the parallelepiped formed by the three basis vectors in B.

Example Given that $A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}, B = \begin{bmatrix} 2 & 5 \\ 2 & 5 \end{bmatrix}$ and $C = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix}$:

$$\begin{aligned} \det(A) &= 1 \times 4 - 2 \times 3 \\ &= -2 \end{aligned}$$

$$\det(B) = 0 \tag{2.2}$$

Equation (2.2) shows a more general property, given that $X = \begin{bmatrix} a & b \\ a & b \end{bmatrix}$, $\det(X) = 0$. The proof is trivial and shall be left as a verification to the reader.

$$\begin{aligned} \det(C) &= 1\begin{vmatrix} 5 & 6 \\ 8 & 9 \end{vmatrix} - 2\begin{vmatrix} 4 & 6 \\ 7 & 9 \end{vmatrix} + 3\begin{vmatrix} 4 & 5 \\ 7 & 8 \end{vmatrix} \\ &= (5 \times 9 - 8 \times 6) - 2(4 \times 9 - 6 \times 7) + 3(4 \times 8 - 7 \times 5) \\ &= 0 \end{aligned}$$

Next, we would move onto the *inverse* of a matrix. Think of a number x, $x^{-1} \times x = 1$, 1 is also called the multiplicative identity. Likewise, given a matrix Y, $YY^{-1} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$ which is the identity matrix. If you think of Y as a transformation, then Y^{-1} "undo" the transformation of Y.

Definition 2.8 The (multiplicative) inverse of a matrix X is denoted X^{-1} where:

$$XX^{-1} = I$$

where I is the identity matrix. If X has an inverse, it is called an *invertible* matrix. X has an inverse if and only if $\det(X) \neq 0$. To find the inverse of a matrix $Y = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$, we use the following formula:

$$Y^{-1} = \frac{1}{\det(Y)} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix} \tag{2.3}$$

When multiplying a matrix by a constant, you multiply each entry in matrix by the constant.

Example

Given $A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$ and $B = \begin{bmatrix} 2 & 5 \\ 2 & 5 \end{bmatrix}$:

$$\begin{aligned} A^{-1} &= \frac{1}{-2}\begin{bmatrix} 4 & -2 \\ -3 & 1 \end{bmatrix} \\ &= \begin{bmatrix} -2 & 1 \\ 1.5 & -0.5 \end{bmatrix} \end{aligned}$$

$\det(B) = 0$, hence B is not invertible.

Until now, this chapter has been focusing mainly on computation. The reader may feel that these computations are excessively dull and boring. That being said, here is a application of determinants, *Cramer's rule*. Cramer's rule may be used to solve a system of linear equations. Firstly, we will represent a simultaneous equation as a product of vectors and matrices. Given a simultaneous equation:

$$\begin{aligned} x + y &= 10 \\ 2x - y &= 14 \end{aligned} \tag{2.4}$$

We write it as:

$$\begin{bmatrix} 1 & 1 \\ 2 & -1 \end{bmatrix}\begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 10 \\ 14 \end{bmatrix} \tag{2.5}$$

This is also called the "matrix form" of a simultaneous equation. It also important to recognize that Equation (2.5) is in the form of $A\vec{v} = B$, it is not easy to see that $\vec{v} = BA^{-1}$. From Equation (2.3), we can find BA^{-1}

$$\begin{aligned} BA^{-1} &= \begin{bmatrix} 10 \\ 14 \end{bmatrix}\begin{bmatrix} \frac{1}{3} & \frac{1}{3} \\ \frac{2}{3} & -\frac{1}{3} \end{bmatrix} \\ &= \begin{bmatrix} 8 \\ 2 \end{bmatrix} \end{aligned}$$

This is also the solution for Equation (2.5), x is indeed 8 and y is indeed 2. However, this method might not be the best. If there were three unknowns and three equations, how could you solve the equations? We may apply Cramer's rule. Consider a system of n linear equations with n unknowns $(x_1, x_2, \dots, x_n)$, it can be represented as $Ax = B$ as we have seen before. The solutions for the equations are:

$$x_i = \frac{\det(A_i)}{\det(A)} \quad i = 1, \dots, n$$

Where x_i is the i-th solution and A_i is the matrix formed by replacing the i-th column of A by B.

Example

Given the following system of linear equations of three unknowns:

$$\begin{aligned} x_1 + x_2 + x_3 &= 12 \\ x_1 - x_2 + x_3 &= 4 \\ x_1 - x_2 - x_3 &= -6 \end{aligned}$$

and its matrix form is:

$$\begin{bmatrix} 1 & 1 & 1 \\ 1 & -1 & 1 \\ 1 & -1 & -1 \end{bmatrix}\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 12 \\ 4 \\ -6 \end{bmatrix}$$

The solutions are:

$$\begin{aligned}
x_1 &= \frac{\det(A_1)}{\det(A)} \\
&= \frac{\det\left(\begin{bmatrix} 12 & 1 & 1 \\ 4 & -1 & 1 \\ -6 & -1 & -1 \end{bmatrix}\right)}{\det\left(\begin{bmatrix} 1 & 1 & 1 \\ 1 & -1 & 1 \\ 1 & -1 & -1 \end{bmatrix}\right)} \\
&= \frac{12}{4} \\
&= 3
\end{aligned}$$

$$\begin{aligned}
x_2 &= \frac{\det(A_2)}{\det(A)} \\
&= \frac{\det\left(\begin{bmatrix} 1 & 12 & 1 \\ 1 & 4 & 1 \\ 1 & -6 & -1 \end{bmatrix}\right)}{\det\left(\begin{bmatrix} 1 & 1 & 1 \\ 1 & -1 & 1 \\ 1 & -1 & -1 \end{bmatrix}\right)} \\
&= \frac{16}{4} \\
&= 4
\end{aligned}$$

$$\begin{aligned}
x_3 &= \frac{\det(A_3)}{\det(A)} \\
&= \frac{\det\left(\begin{bmatrix} 1 & 1 & 12 \\ 1 & -1 & 4 \\ 1 & -1 & -6 \end{bmatrix}\right)}{\det\left(\begin{bmatrix} 1 & 1 & 1 \\ 1 & -1 & 1 \\ 1 & -1 & -1 \end{bmatrix}\right)} \\
&= \frac{20}{4} \\
&= 5
\end{aligned}$$

As you can tell, Cramer's rule is very powerful. In the real world, however, *gaussian elimination* is a much faster alternative than Cramer's rule. It is also unlikely for the reader to be computing large matrices determinants by hand, we have computers for that. This also applies to matrix multiplication. This book, and more generally calculus courses only need knowledge on computing determinants of up to 3×3 matrices.

Exercise

1. Find the determinants and inverses, if any, for the following matrices:

   (a) $\begin{bmatrix} 1 & 9 \\ 2 & 10 \end{bmatrix}$

   (b) $\begin{bmatrix} 3 & 2 \\ 6 & 4 \end{bmatrix}$

(c) $\begin{bmatrix} 3 & 4 \\ 7 & 9 \end{bmatrix}$

2. Find the determinant of the following matrix:

$$\begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & 4 \\ 2 & 3 & 6 \end{bmatrix}$$

3. Solve the following simultaneous equations using matrices:

$$\begin{aligned} x_1 + x_2 + x_3 &= 3 \\ x_1 + 2x_2 + x_3 &= 4 \\ x_1 + 2x_2 + 2x_3 &= 6 \end{aligned}$$

Solution

1. (a) -8 and $\begin{bmatrix} -\frac{10}{8} & \frac{9}{8} \\ \frac{1}{4} & -\frac{1}{8} \end{bmatrix}$
   (b) 0 and no inverse
   (c) -1 and $\begin{bmatrix} -9 & 4 \\ 7 & -3 \end{bmatrix}$

2. 1

3. Use Cramer's rule and $x_1 = 0$, $x_2 = 1$, $x_3 = 2$. Example for x_1:

$$\begin{aligned} x_1 &= \frac{\det(A_1)}{\det(A)} \\ &= \frac{\begin{vmatrix} 3 & 1 & 1 \\ 4 & 2 & 1 \\ 6 & 2 & 2 \end{vmatrix}}{\begin{vmatrix} 1 & 1 & 1 \\ 1 & 2 & 1 \\ 1 & 2 & 2 \end{vmatrix}} \\ &= \frac{0}{1} \\ &= 0 \end{aligned}$$

Example for x_2:

$$\begin{aligned} x_2 &= \frac{\det(A_2)}{\det(A)} \\ &= \frac{\begin{vmatrix} 1 & 3 & 1 \\ 1 & 4 & 1 \\ 1 & 6 & 2 \end{vmatrix}}{\begin{vmatrix} 1 & 1 & 1 \\ 1 & 2 & 1 \\ 1 & 2 & 2 \end{vmatrix}} \\ &= \frac{1}{1} \\ &= 1 \end{aligned}$$

SECTION 2.5

Cross products and triple products

The previous section perfectly segues into this section. *Cross products* is an operation on vectors that happens solely in the third dimension. Do not confuse it with the dot product, both uses different operators and should not be used interchangeably. Firstly, we must recall that one of the basis vectors is $\hat{\mathbf{k}}$ and it is one unit in the z direction. The cross product is a vector, whose magnitude is the area of a parallelogram formed by the two operands(which are vectors). More formally, the cross product is defined as:

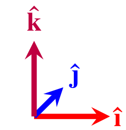


Figure 2.16. Figure of the three basis vectors.

Definition 2.9 The *cross product* is an operation on two vectors and the result is another vector which is perpendicular to both vectors. It is defined by the formula:

$$\vec{a} \times \vec{b} = (||\vec{a}||)(||\vec{b}||)\sin(\theta)\vec{n}$$

where θ is the angle between $\vec{a}$ and $\vec{b}$ and $\vec{n}$ is the unit vector perpendicular to both of the operands such that all three vectors obey the "right hand rule". The common way to compute the cross product is using the following formula:

$$\vec{a} \times \vec{b} = \begin{vmatrix} \hat{\mathbf{i}} & \hat{\mathbf{j}} & \hat{\mathbf{k}} \\ \vec{a}_x & \vec{a}_y & \vec{a}_z \\ \vec{b}_x & \vec{b}_y & \vec{b}_z \end{vmatrix} \tag{2.6}$$

Lastly, do not confuse $\vec{n}$ with $\hat{\mathbf{k}}$ as both vectors are sometimes not the same.

The operands of $\vec{a} \times \vec{b}$ are $\vec{a}$ and $\vec{b}$.

The right hand rule states that if you use your right hand and point the index finger in the direction of $\vec{a}$, your middle finger in the direction of $\vec{b}$, your thumb should be pointing in the direction of $\vec{n}$.

The cross product is very ubiquitous in physics and vector calculus alike.

Example Given that $\vec{a} = \begin{bmatrix} 1 \\ 2 \\ 0 \end{bmatrix}$ and $\vec{b} = \begin{bmatrix} -2 \\ 1 \\ 0 \end{bmatrix}$, then $\vec{a} \times \vec{b}$ would be:

$$\begin{aligned} \vec{a} \times \vec{b} &= \begin{vmatrix} \hat{\mathbf{i}} & \hat{\mathbf{j}} & \hat{\mathbf{k}} \\ 1 & 2 & 0 \\ -2 & 1 & 0 \end{vmatrix} \\ &= 0\hat{\mathbf{i}} - 0\hat{\mathbf{j}} + 5\hat{\mathbf{k}} \\ &= 5\hat{\mathbf{k}} \end{aligned}$$

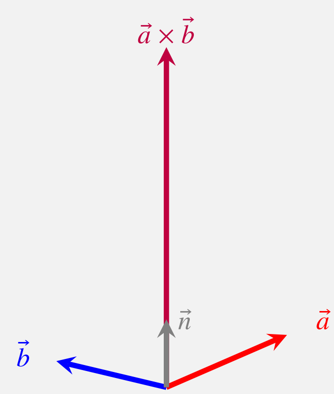


Figure 2.17. Figure of two vectors and their cross product.

An important fact to note is that the operation is not commutative. In fact, the operation is called *anti-commutative*. This means that $\vec{a} \times \vec{b} = -\vec{b} \times \vec{a}$. In addition, if the angle between the operands is any integer multiple of π, then the result is $\vec{0}$. This should be trivially seen from Definition 2.9. The derivation for the cross product is also easy, given two vectors $\vec{a}$ and $\vec{b}$ and a angle θ between them, simply use the sine rule and the result should be obvious. The cross product has many real world applications. Take angular momentum for example, it is given by $L = r \times p$, where r is the position vector and p is the linear momentum of a particle. Finally, always remember that cross products are only in the third dimension and they use the $\times$ operator.

Next, we will move on to triple products which happens to also be a three dimensional topic. Firstly, take a look at the following three vectors:

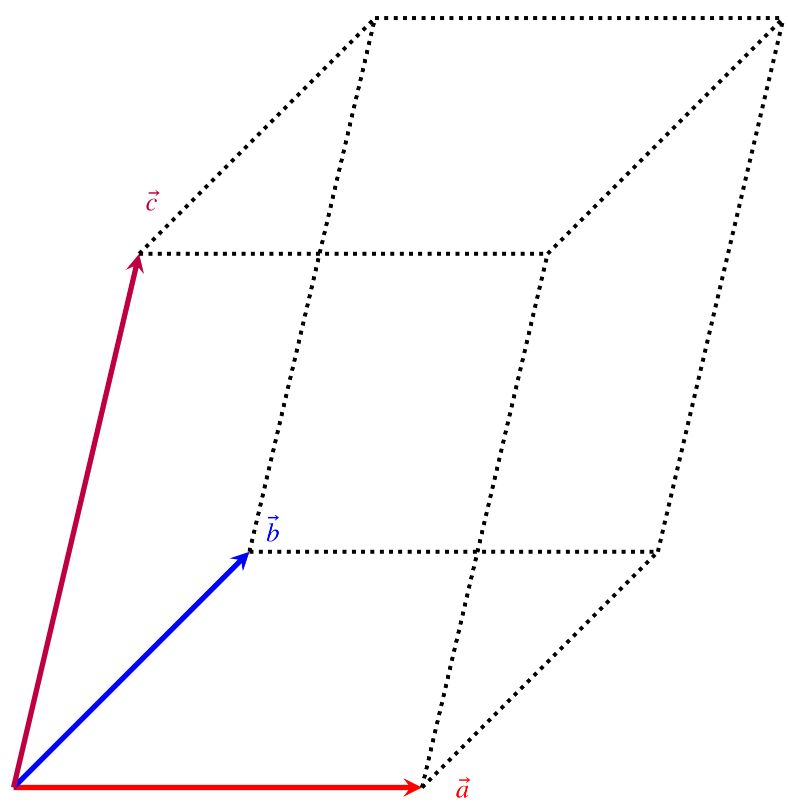


Figure 2.18. Figure of triple vector multiplication.

Figure 2.18 is the example of triple vector multiplication. You would represent it as $\vec{a}\cdot(\vec{b}\times\vec{c})$. The result here is simply the volume of the parallelepiped formed in Figure 2.18.

Definition 2.10 The *triple vector product* is a product of three 3 dimensional vectors. The most common of which, is the scalar triple product which is Figure 2.18. The formula for which is $\vec{a} \cdot (\vec{b} \times \vec{c})$, which yields the volume of the parallelepiped with the sides being the three vectors. Another type of triple vector multiplication is vector triple product. It is simply $\vec{a} \times (\vec{b} \times \vec{c})$. We can use *Lagrange's formula* to compute it easily: $\vec{a} \times (\vec{b} \times \vec{c}) = \vec{b}(\vec{a} \cdot \vec{c}) - \vec{c}(\vec{a} \cdot \vec{b})$.

Example We shall derive the formula for the volume of a parallelepiped given its height, breadth and length. We shall denote them h, b and l respectively. Notice that we define the sides of the parallelepiped like this:

$$\vec{a} = \begin{bmatrix} l \\ 0 \\ 0 \end{bmatrix}$$
$$\vec{b} = \begin{bmatrix} 0 \\ b \\ 0 \end{bmatrix}$$
$$\vec{c} = \begin{bmatrix} 0 \\ 0 \\ h \end{bmatrix}$$

Now, we use the triple vector multiplication formula. However, take note of the brackets and the order of operations:

$$\begin{aligned} \vec{a} \cdot (\vec{b} \times \vec{c}) &= \vec{a} \cdot \begin{vmatrix} \hat{\mathbf{i}} & \hat{\mathbf{j}} & \hat{\mathbf{k}} \\ 0 & b & 0 \\ 0 & 0 & h \end{vmatrix} \\ &= \begin{bmatrix} l \\ 0 \\ 0 \end{bmatrix} \cdot \begin{bmatrix} bh \\ 0 \\ 0 \end{bmatrix} \\ &= lbh \end{aligned} \tag{2.7}$$

The result in Equation (2.7) is trivial and should come as no surprise to the reader.

As for vector triple product, the process of computation is very similar. We first evaluate the cross product of the two vectors in parenthesis, then take the cross product of the first and resultant vector from the previous computation. We shall give a simple proof for Lagrange's formula.

Theorem 2.1 (*Lagrange's formula*) Given $\vec{a}, \vec{b}, \vec{c}$, show that $\vec{a} \times (\vec{b} \times \vec{c}) = \vec{b}(\vec{a} \cdot \vec{c}) - \vec{c}(\vec{a} \cdot \vec{b})$.

A sidenote here, when it is written $\vec{a}\vec{b}$, this is not the cross or dot product, instead it is equal to $\begin{bmatrix} a_x b_x \\ a_y b_y \\ \cdots \end{bmatrix}$.

Proof We will start by evaluating $\vec{a} \times (\vec{b} \times \vec{c})$:

$$\begin{aligned} \vec{a} \times (\vec{b} \times \vec{c}) &= \vec{a} \times \begin{vmatrix} \hat{\mathbf{i}} & \hat{\mathbf{j}} & \hat{\mathbf{k}} \\ b_x & b_y & b_z \\ c_x & c_y & c_z \end{vmatrix} \\ &= \vec{a} \times \begin{bmatrix} b_y c_z - b_z c_y \\ b_z c_x - b_x c_z \\ b_x c_y - b_y c_x \end{bmatrix} \\ &= \begin{vmatrix} \hat{\mathbf{i}} & \hat{\mathbf{j}} & \hat{\mathbf{k}} \\ a_x & a_y & a_z \\ b_y c_z - b_z c_y & b_z c_x - b_x c_z & b_x c_y - b_y c_x \end{vmatrix} \\ &= \begin{bmatrix} a_y(b_x c_y - b_y c_x) - a_z(b_z c_x - b_x c_z) \\ \cdots \end{bmatrix} \\ &= \begin{bmatrix} a_y b_x c_y + a_z b_x c_z - a_y b_y c_x - a_z b_z c_x \\ \cdots \end{bmatrix} \end{aligned} \tag{2.8}$$

We will examine the $\hat{\mathbf{i}}$ component of 2.8, the other two components are similar. Then, notice that:

$$\begin{aligned}\vec{b}(\vec{a}\cdot\vec{c})-\vec{c}(\vec{a}\cdot\vec{b}) &= \vec{b}(a_xc_x+a_yc_y+a_zc_z)-\vec{c}(a_xb_x+a_yb_y+a_zb_z)\\ &= \begin{bmatrix} b_x(a_xc_x+a_yc_y+a_zc_z)\\ b_y(a_xc_x+a_yc_y+a_zc_z)\\ b_z(a_xc_x+a_yc_y+a_zc_z)\end{bmatrix}-\begin{bmatrix} c_x(a_xb_x+a_yb_y+a_zb_z)\\ c_y(a_xb_x+a_yb_y+a_zb_z)\\ c_z(a_xb_x+a_yb_y+a_zb_z)\end{bmatrix}\\ &= \begin{bmatrix} a_yb_xc_y+a_zb_xc_z-a_yb_yc_x-a_zb_zc_x\\ \cdots\end{bmatrix} \qquad (2.9)\end{aligned}$$

As you can see, the $\hat{\mathbf{i}}$ components of 2.8 and 2.9 are identical. Hence, we have shown Lagrange's formula. QED

Remark The above proof is a tedious pedagogical exercise mainly created to drill in the concepts of the cross products.

Exercise

1. Find the cross products of the following vectors:
   (a) $\begin{bmatrix}1\\2\\5\end{bmatrix}\times\begin{bmatrix}1\\2\\7\end{bmatrix}$
   (b) $\begin{bmatrix}0\\2\\4\end{bmatrix}\times\begin{bmatrix}1\\3\\5\end{bmatrix}$
2. Prove or disprove:
   (a) $\vec{a}\times\vec{b}=\vec{b}\times\vec{a}$
   (b) $\vec{a}\times\vec{a}=0$
3. Prove Lagrange's formula for the other two components not shown in the previous proof.

Solution

1. (a) $4\hat{\mathbf{i}}-2\hat{\mathbf{j}}$
   (b) $-2\hat{\mathbf{i}}+4\hat{\mathbf{j}}-2\hat{\mathbf{k}}$
2. (a) *Disprove*. Notice that while in question 1(a), the answer is $4\hat{\mathbf{i}}-2\hat{\mathbf{j}}$, if you swap the two vectors, you instead get $-4\hat{\mathbf{i}}+2\hat{\mathbf{j}}$, which is the anti-commutative property of cross products.
   (b) *Prove*. This is a trivial result by using the formula for cross products.
3. This is also a trivial and tedious result that can be shown just by some evaluation of the cross products.

Remark While the linear algebra chapter in this book is ending, this is a very small subset of a standard linear algebra course. Many concepts, such as eigenvalues, are not covered due to their not so important nature. Readers are still highly encouraged to read up some of the other concepts.

Section 2.6
An aside: Duality

TBD

PART

II

Calculus I

CHAPTER 3

Derivatives

SECTION 3.1

The definition of derivatives

Onto the calculus portion that is very much anticipated. No more the quick skimping over precalculus topics. We shall enter this discussion with a very simple question that will yield many interesting results. Take a look at the graph below:

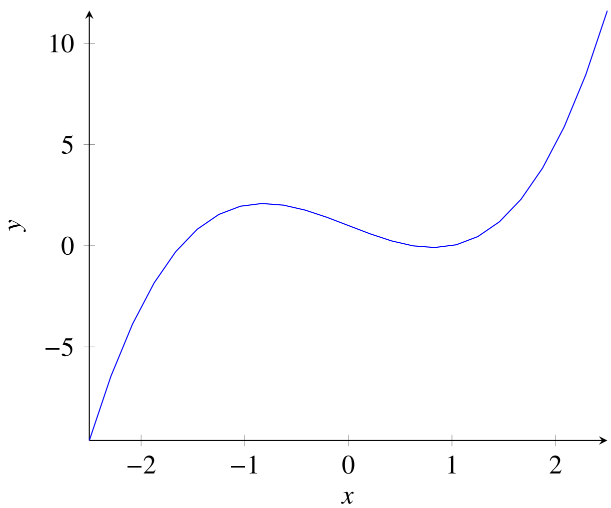


Figure 3.1. Graph of $f(x) = x^3 - 2x + 1$.

From Figure 3.1, we see that this is a cubic graph. It should not to be hard to see that the graph is always changing value. Also, it should not be difficult to see that the rate of change of the graph is not constant. We can use the gradient formula to verify this:

Definition 3.1 The gradient between two points $(x, f(x))$ and $(x + a, f(x + a))$ is:

$$\textit{Gradient} = \frac{\Delta y}{\Delta x} = \frac{f(x+a) - f(x)}{a} \tag{3.1}$$

Using Definition 3.1, we can pick some points on the graph. You would notice that if you plug in different points then the gradient is different. The main, central question is: if the rate of change is not constant, then what is the rate of change at a single point along the function? This is where the gradient formula fails. We want something that gives us the gradient of the graph at every single point along the graph. The forefathers of calculus were all brilliant and they noticed a special fact that applies to many functions. Let us zoom in to the graph where $x = 2$, we want to zoom in as closely as we can.

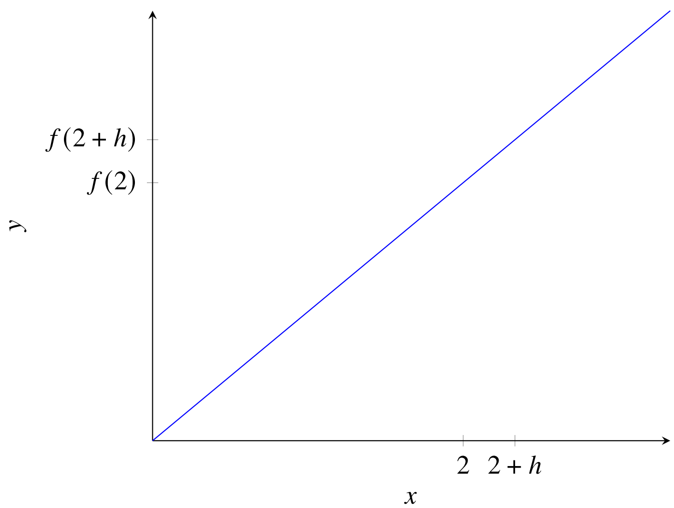


Figure 3.2. Zoomed in graph of Figure 3.1.

In Figure 3.2, we realise that the graph, when zoomed in sufficiently, essentially is a straight line. This is exciting, we can use our gradient formula which works on straight lines. In addition, on the x axis, we mark a $2+h$, a small distance away from 2. We can infer that h is a really small number, almost going to 0 as we keep zooming in. We use the limit notation to denote this. Hence, we can denote it as $\lim_{h\to 0}$. You may be itching to use the gradient formula now. We can substitute in the values:

The limit notation and its concepts will be discussed in a later chapter, for now, just keep in mind that this means h is a really tiny number, almost 0.

$$\lim_{h\to 0} \frac{f(2+h)-f(2)}{h}$$

However, notice that we have to divide throughout by h, and h is very small, infinitesimally small. Some careless manipulation and you would say the expression blows up to infinity. What the formula does tell us, though, is the gradient practically at 2. This gives us an idea. Can we just replace 2 with any value of x? This brings us to the definition of the *derivative*.

Definition 3.2 The *derivative* of a function $f(x)$ is the instantaneous rate of change of that function. It can be written as $f'(x)$, $\frac{df(x)}{dx}$ or $\dot{f}$.

$$f'(x) = \lim_{h\to 0} \frac{f(x+h)-f(x)}{h} \tag{3.2}$$

Sometimes, to save writing a few letters, the inputs to the functions may be omitted, such as f instead of $f(x)$. This is more common when the function takes in multiple inputs.

In the definition of the derivative, we can see that the derivative is also a function. Also, the definition states that it is the "instantaneous rate of change". This simply means that we are looking at the rate of change of the function over a very, very tiny range of x. Of course, the definition may feel contradictory. How can some value change "instantaneously"? Well, this is resolved by not exactly by substituting $\Delta y = 0$, instead we let it "approach" 0. As a result, using this clever little mathematical trick, we avoided this problem. However, some of you may have noticed we have not evaluated any derivatives, we just defined them. It is no good to not answer our original question. To illustrate a derivation of a derivative, we shall use a much simpler graph, $f(x) = x^2$.

From Figure 3.3, we can consider what happens at $x = 2$. We shall use the definition of the derivatives, which is also more commonly referred to as "first principles".

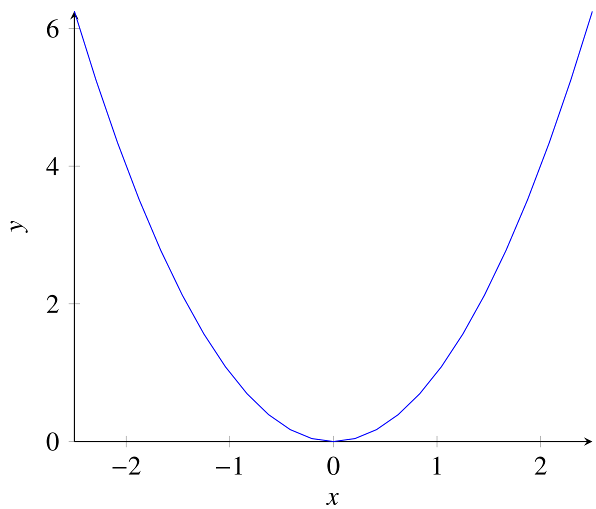


Figure 3.3. Figure of $y = x^2$.

Example

To find the derivative at $x = 2$, we use the first principles as such:

$$\begin{aligned} f'(2) &= \lim_{h\to 0} \frac{f(2+h)-f(2)}{h} \\ &= \lim_{h\to 0} \frac{(2+h)^2-4}{h} \\ &= \lim_{h\to 0} \frac{4+4h+h^2-4}{h} \\ &= \lim_{h\to 0} 4+h \\ &= 4 \end{aligned}$$

To find the derivative of f, we use a similar manipulation:

$$\begin{aligned} f'(x) &= \lim_{h\to 0} \frac{f(x+h)-f(x)}{h} \\ &= \lim_{h\to 0} \frac{x^2+2xh+h^2-x^2}{h} \\ &= \lim_{h\to 0} 2x+h \\ &= 2x \end{aligned}$$

We can ignore h in the second last step because as a general rule of thumb, as long as substituting $h = 0$ does not create an expression of $\frac{a}{0}$, we can let $h = 0$. Of course, this is very not rigorous and there are other exceptions that will be very evident after the reader has read the later chapter of limits.

In other words, we have also found a function of the gradient of the tangent line for any point along $f(x) = x^2$. Lastly, I would like to comment that normally, if not explicitly stated in the question to use "first principles", you can use the derivative rules you will learn in the next chapter to solve derivatives much more quickly and efficiently. In all honesty, using the "first principles" is very tedious and generally would not be used.

Exercise

1. Using first principles, find the derivatives of the following functions:
   (a) $f(x) = x^2 + 1$
   (b) $g(x) = x^3 + x$
   (c) $h(x) = \frac{1}{x}$
2. Find the derivative of $\sin(x)$. (Hint: $sin(a+b) = sin(a)cos(b) + cos(a)sin(b)$).

3. **(Difficult)** Let $f(x) = (x-1)(x-2)\cdots(x-1729)$, find $f'(1729)$. *(Adapted from SMO 2025 Senior Section Round 1).*